

Assignment

- 1) The workfunction of the potassium is calculated to be 1.9 eV , when ultraviolet light of wavelength $3600 \text{ \AA}$ falls on the potassium surface. To find max K.E of emitted photoelectrons.

Sol

$$KE = \left[\text{Energy of incident photons} \right] - \left[\text{Workfunction} \right]$$

Energy of incident photons $= E = hf$

h (Planck's constant) $\approx 6.626 \times 10^{-34} \text{ J/s}$

c (speed of light) $= 3 \times 10^8 \text{ m/s}$

λ (wavelength) $= 3600 \text{ \AA} = 3600 \times 10^{-10} \text{ m}$

$$E = 6.626 \times 10^{-34} \text{ J/s} \times \frac{3 \times 10^8 \text{ m/s}}{3600 \times 10^{-10} \text{ m}}$$

$$E = 5.51 \text{ eV}$$

$$E = 5.51 \text{ eV}$$

$$\Rightarrow \text{Maximum KE} = 5.51 \text{ eV} - 1.9 \text{ eV}$$

$$KE_{\text{max}} = 3.61 \text{ eV}$$

$\Rightarrow$ Velocity of ejected electron using KE:

$$KE = \frac{1}{2} mv^2$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$3.61 \text{ eV} = \left(\frac{1}{2} \right) (9.11 \times 10^{-31} \text{ kg}) \times v^2$$

Solving for v :

$$v^2 \approx (2 \times 3.61 \text{ eV}) / (9.11 \times 10^{-31} \text{ kg})$$

$$v = 1.062 \times 10^6 \text{ m/s}$$

∴ The maximum KE of emitted photoelectrons is approximately 3.61 eV, and the velocity of the ejected electron is approximately 1.62×10^6 m/s.

2) Sol.

The energy of the photon emitted when the proton undergoes a transition from the first excited state ($n=2$) to the ground state ($n=1$) can be calculated as:

$$E = \frac{(h^2 \times n^2)}{8 m L^2}$$

L = length of box.

h = Planck's constant

$n =$

$$E = \frac{(6.626 \times 10^{-34} \text{ J/s})^2 \times (2^2)}{8 (1.67 \times 10^{-27} \text{ kg}) (10 \times 10^{-15} \text{ m})^2}$$

$$E = 4.99 \times 10^{-13} \text{ J}$$

To calculate wavelength of a photon, using energy-wavelength relationship:

$$E = h (c/\lambda) \quad , \quad c \text{ is the speed of light}$$

$$\lambda = \frac{(6.626 \times 10^{-34} \text{ J/s}) \times (3 \times 10^8 \text{ m/s})}{(4.99 \times 10^{-13} \text{ J})}$$

$$\lambda = 3.98 \times 10^{-15} \text{ m}$$

→ belongs to gamma-ray-region

3) SolTo find : $KE_{\max}$ 1) Maximum Kinetic Energy [$KE_{\max}$]:

$$KE_{\max} = hf - \phi \rightarrow \text{workfunction of Cerium metal (2.14 eV)}$$

Convert workfunction to joules:-

$$\phi = 2.14 \text{ eV} \times 1.602 \times 10^{-19} \text{ J/eV}$$

Now, Calculate $KE_{\max}$:

$$KE_{\max} = \left(\frac{1}{2}\right) m v_{\max}^2$$

 m = mass of electron ($9.11 \times 10^{-31} \text{ kg}$ approx.)

$$\Rightarrow v_{\max}^2 = (2 \times KE_{\max}) / m$$

$$v_{\max}^2 = \frac{(2 \times 3.98 \times 10^{-19} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}$$

$$v_{\max} = 6.64 \times 10^5 \text{ m/s}$$

$\therefore$ Maximum Kinetic Energy of the emitted electrons is approximately $3.98 \times 10^{-19} \text{ J}$,
 $\rightarrow$ Maximum velocity of the emitted photoelectrons is approximately $6.64 \times 10^5 \text{ m/s}$.

4)

Solution:* Photoelectric Effect:-

It is a phenomenon in physics where electrons are emitted from the surface of the metal when it is exposed to electromagnetic radiation such as light.

⇒ Threshold Wavelength (λ)

* Minimum Energy required to release an electron from the metal surface, which is equal to the workfunction (ϕ).

$$E_{\text{threshold}} = \phi \quad \text{--- (1)}$$

Energy of a photon can be calculated by the formula:

$$E_{\text{photon}} = \frac{hc}{\lambda} \quad \text{--- (2)}$$

Now, Equating Equation (1) = (2)

$$\frac{hc}{\lambda} = \phi$$

$$\lambda_{\text{threshold}} = \frac{hc}{\phi}$$

$$\lambda_{\text{threshold}} = \frac{(6.626 \times 10^{-34} \text{ J.s}) (3 \times 10^8 \text{ m/s})}{(8.01 \times 10^{-19})}$$

$$\lambda_{\text{threshold}} = 2.48 \times 10^{-7} \text{ m} \rightarrow 248 \text{ nm}$$

→ Energy of workfunction in eV :
 $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

$$E_{\text{workfunction}} (\text{eV}) = \frac{\phi}{1.602 \times 10^{-19} \text{ J/eV}}$$

Substituting the values of ϕ in

$$E_{\text{workfunction}} (\text{eV}) = \frac{\phi}{1.602 \times 10^{-19} \text{ J/eV}}, \text{ we get:}$$

$$E_{\text{workfunction}} (\text{eV}) = \frac{8.01 \times 10^{-19} \text{ J}}{1.602 \times 10^{-19} \text{ J/eV}}$$

$$E_{\text{workfunction}} = 5 \text{ eV}$$

∴ The energy of the workfunction is approximately 5 eV.

5) Essential condition for quantization:

* The essential condition for quantization is the confinement or restriction of a physical quantity, such as energy or angular momentum, to discrete, quantized values or levels. This occurs when the system or environment imposes constraints that restrict the allowed values, leading to quantization in quantum mechanics, as seen in energy levels of electrons in atoms.

6) Eigenfunction:-

* It is a special function that, when operated upon by a mathematical operator, returns a scaled version of itself.

* It is a function that doesn't change its shape when a particular operation is applied to it.

* These functions used to describe the possible states of a physical system, like wavefunctions of electrons in an atom.

Eigenvalue:-

* It is a corresponding scale factor that tell us how much the eigenfunction is stretched or shrunk when the mathematical operator acts on it.

* It represents a physical property or characteristic value of the system.

* These values often represent observable quantities like energy levels or angular momentum.

7) Node:-

* A Node refers to a point or region in a quantum mechanical wavefunction where the probability of find a particle is zero.

i) In first quantum state, or ground state, the ground state of a particle in a 1D box has no nodes. It is described by a single sine wave and is the

lowest energy state the particle can occupy.

ii) In second quantum state, the first excited state of a particle in 1D box has one node. It is described by a single sine wave with one point where the wavefunction passes through zero.

8. Wave Function :-

* It is a mathematical expression that describes the behaviour and properties of particles, and it is represented by the symbol " ψ ".

* It provides information about the probability distribution in various states and positions.

⇒ Properties of Wave Function:

* The properties of wave function are: probability, Amplitude, Continuity, Complex, Normalisation, Superposition, stationary, Eigenfunction and orthogonal.

9) Normalization:-

Normalization is a process in quantum mechanics that ensures the total probability of finding a particle in all possible states equals 1.

Mathematically, for a wavefunction $\psi(x)$, normalization is represented as:

$$\int |\psi(x)|^2 dx = 1$$

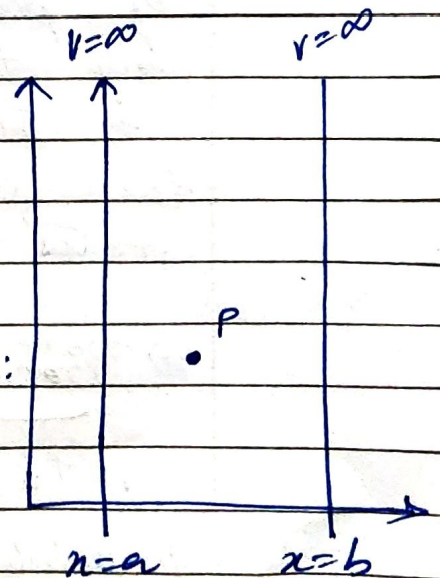
10) Schrodinger equation for particle in 1D box.

* Consider that p is a particle inside an infinitely long enclosed box with the particle.

* PE is the max ~~to~~ at the corners and it tends up to infinity.

Mathematically, it is expressed as:

$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$



Now, from kinetic energy operator:-

$$\frac{-\hbar^2}{2m} \text{KE} \psi(x) = \left(\frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \psi(x)$$

The kinetic energy operator is given by:

$$\text{KE} = \frac{-\hbar^2}{2m} \frac{d^2}{dx^2}$$

11) Q

For one-dimensional box of length L , the normalised wave functions for the first three electronic states are given by:-

In Ground state; at $n=1$;

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

In First Excited state; at $n=2$;

$$\psi_2(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right)$$

In Second excited state; at $n=3$

$$\psi_3(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi x}{L}\right)$$

To compute the energy of the third states $n=3$, you can use the formula for the energy levels in a 1D box.

$$E_n = \frac{n^2 h^2 \pi^2}{2mL^2}$$

By, substituting $n=3$ in the equation, we will get:

$$E_3 = \frac{(3)^2 h^2 \pi^2}{2mL^2}$$

$$E_3 = \frac{9 h^2 \pi^2}{2mL^2}$$

12) the physical interpretations of quantum numbers m , l , and m_l in quantum mechanics:

1. Magnetic Quantum number (m) specifies the orientations of an atomic orbital in space.

2. The Azimuthal Quantum number (l) which is also known as the angular momentum quantum number. Describes shape of an orbital.

3. The magnetic quantum number m_l further describes the orbital's orientation within a given l subshell.

* These quantum numbers help describe the electron's distribution and behaviour within an atom.

13) * We use polar coordinates to solve hydrogen atom because, these are used to solve the hydrogen atom because they are naturally suited to the spherical symmetry of the problem.

* The Electron's motion in the hydrogen atom is influenced by the nucleus at the origin, making spherical coordinates a more intuitive choice to describe and solve the problem efficiently.